

English-Source Interpretation Demand

Voyce data, Jan–Apr 2025 | CIUSSS Centre-Ouest-de-l'Île-de-Montréal

Executive Summary

The headline

English is the dominant source language at this client — **7,288 of 7,625 calls (95.6%)** originate in English, spread across **89 distinct target languages**. The cancellation rate is **15.9%**, a healthy contrast to the 75.7% seen on French-source traffic. Demand is highly concentrated: just three target languages (**Punjabi, Spanish, Bengali**) account for **50%** of all serviced minutes, and ten languages cover 80%. The remaining 79 languages collectively form a long tail.

Key numbers (Jan–Apr 2025)

Metric	Value	Notes
Total platform calls	7,625	4 months
English-source calls	7,288	95.6% of all volume
— Serviced	6,131	22.6 min average duration
— Cancelled	1,157	0-minute (pre-connect)
English-source cancellation rate	15.9%	vs. 75.7% French
Distinct target languages	89	76 with non-zero serviced volume
Annualized true demand	492,814 min/yr	≈ 8,214 hrs/yr
Per-pair interpreters needed	77	one per target language
Shared-pool interpreters needed (true)	7	if pool is fungible

RED FLAG

Video calls cancel 6.7× more than audio

30.7% of English-source video requests were cancelled, versus just **4.6%** for audio — yet 80 of 82 served French calls were also video. This points to a video-routing or first-mile reliability issue worth investigating before scaling video-preferred services.

Recommendation

Size dedicated capacity around the **top 10 target languages** that carry 80% of serviced volume (Punjabi, Spanish, Bengali, Hindi, Arabic, Russian, Tamil, Turkish, Urdu, Chinese Mandarin). Treat the long tail (50+ languages, <5% of volume) as on-demand — a shared 7-interpreter network covers the entire residual load at standard capacity. Shift patterns should weight 9:00–14:00 EST, where 60% of the day's volume occurs.

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1 Scope and Method

1.1 Source data

The dataset is the Voyce export titled “*Voyce data Jan – Apr JEWISH MEMORIAL.xlsx*”, covering the full **CIUSSS Centre-Ouest-de-l’Île-de-Montréal** network: 72 service locations, 4 client identifiers, and 45 distinct service lines. The top-volume locations are CLSC Côte-des-Neiges (1,169 calls), CLSC Parc-Extension (1,109), and PRAIDA Administration (1,008).

1.2 Time window

Coverage	English-source records
January 2025	1,998
February 2025	1,776
March 2025	1,849
April 2025	1,665
Total (4 months)	7,288

All annualized figures apply a $\times 3.0$ multiplier as the file covers four months. The May–December 2025 export should replace the projection with actuals once available.

1.3 Filter applied

Column G (Source Language) filtered to English yields **7,288 rows** across 89 distinct target languages.

1.4 Interpreter capacity assumption

One interpreter is assumed to absorb **6,400 serviced minutes per month** (≈ 25 billable hours/week, accounting for idle time, breaks and ramp-up). A conservative scenario (4,800 min/mo) and an aggressive scenario (8,000 min/mo) are reported in the workbook for sensitivity.

2 Demand Profile

2.1 Volume is concentrated in a handful of languages

Target language	Calls	Min (4mo)	Avg/call	Min/yr
Punjabi	1,462	33,148	22.7	99,444
Spanish	1,101	23,035	20.9	69,105
Bengali	931	18,470	19.8	55,410
Hindi	561	13,623	24.3	40,869
Arabic	299	6,999	23.4	20,997
Russian	181	5,212	28.8	15,636
Tamil	157	3,648	23.2	10,944
Turkish	150	3,314	22.1	9,942
Urdu	125	2,906	23.2	8,718
Chinese Mandarin	120	2,780	23.2	8,340

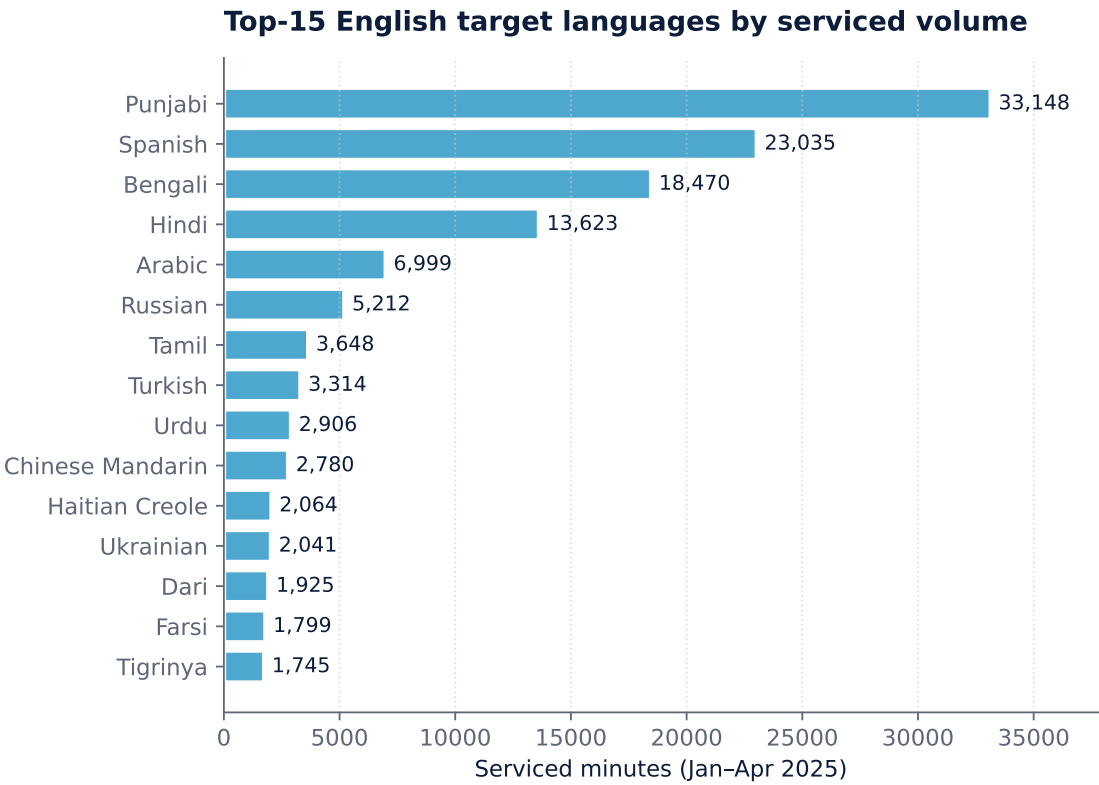


Figure 1: Top-15 target languages by serviced minutes. Punjabi alone represents 24% of all served English-source minutes.

2.2 The long tail accounts for <5%

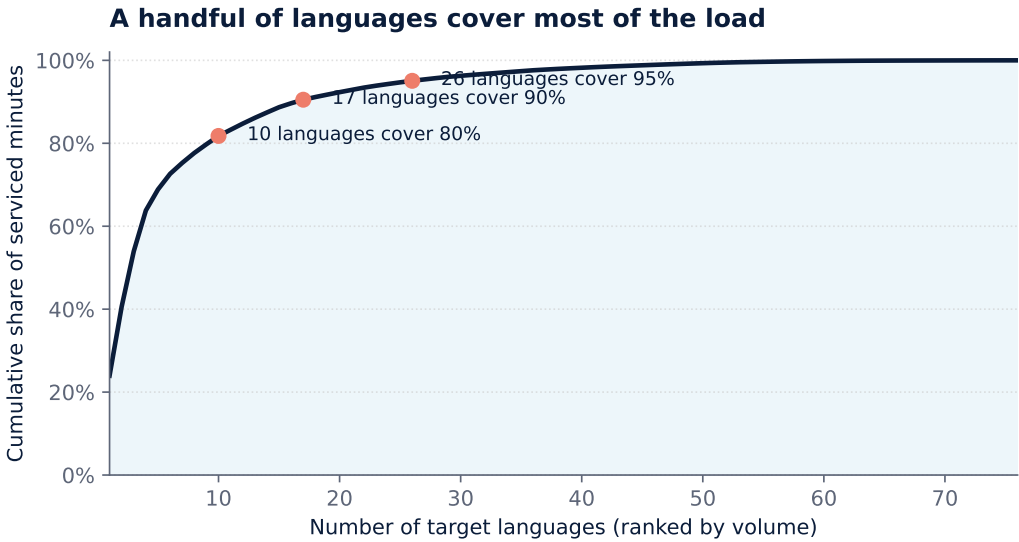


Figure 2: Cumulative share of serviced minutes against languages ranked by volume. The 80/20 rule applies cleanly: 10 languages cover 80% of load, 17 cover 90%, and the bottom 50+ languages together account for under 5%.

What this means for staffing

Dedicated head-count is only justified for the top tier. Languages beyond the top 17 should be sourced from a shared on-demand pool, with SLA tolerances tuned for slightly longer match-time. This is also the profile that benefits most from cross-trained generalists.

2.3 Modality split

Modality	Total	Serviced	Cancelled	Cx rate
Audio	4,133	3,944	189	4.6%
Video	3,148	2,181	967	30.7%
ASL	7	6	1	14.3%

RED FLAG

Video reliability is the single largest unmet-demand driver

Audio cancellations look healthy at 4.6%. Video drops to 30.7% — a $6.7\times$ gap. This single split explains 84% of the cancellations in the dataset (967 of 1,157). Diagnosing video-routing reliability is likely a higher-leverage intervention than hiring additional interpreters.

2.4 Time-of-day pattern

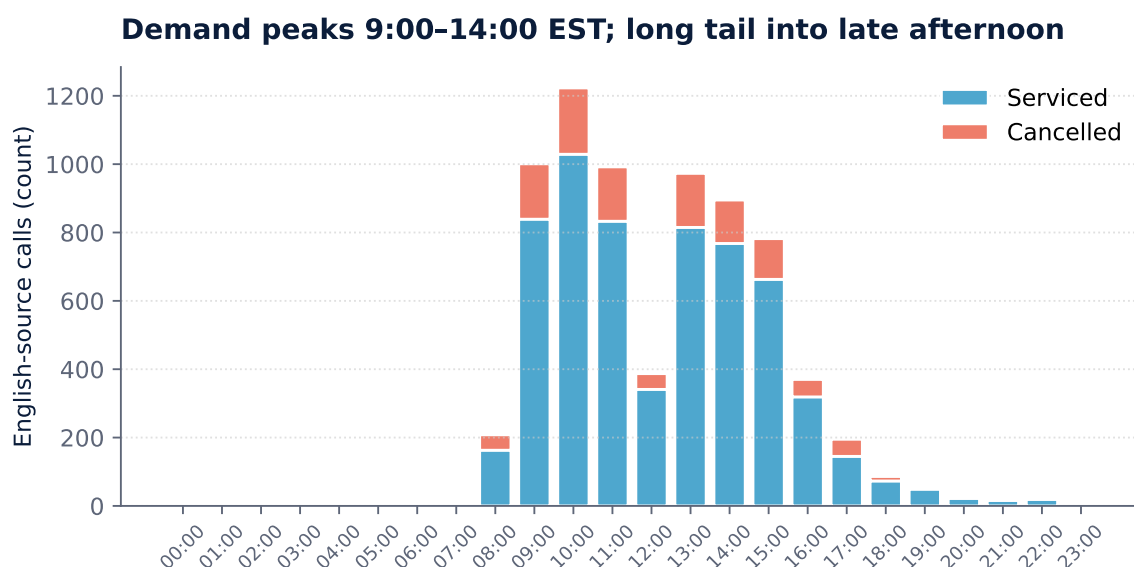


Figure 3: English-source demand peaks in mid-morning and again post-lunch. Cancellations follow demand — absolute counts rise with volume, but rate stays comparatively flat.

3 Cancellation Audit — Outliers and Hot-spots

3.1 Most pairs sit near the 15.9% baseline; a few don't

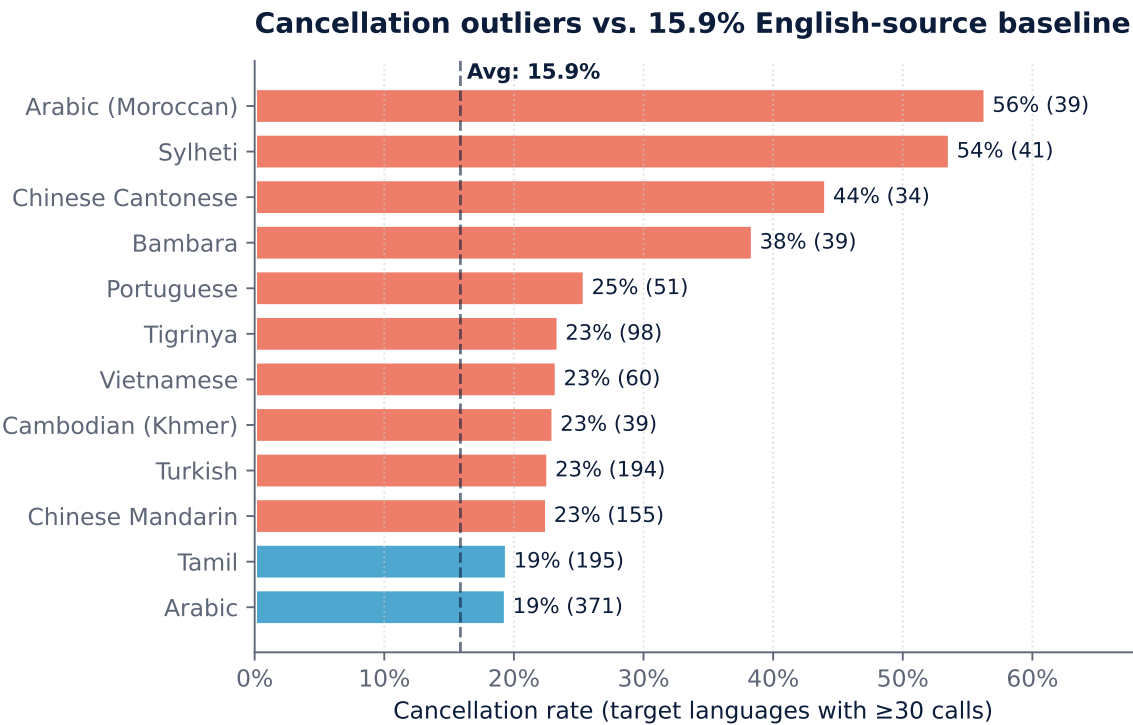


Figure 4: Cancellation rate among target languages with ≥30 calls. The dashed line marks the 15.9% English-source average. Languages above the line warrant capacity review.

Pair	Total calls	Cancelled	Cx rate
English → Arabic (Moroccan)	39	22	56.4%
English → Sylheti	41	22	53.7%
English → Chinese Cantonese	34	15	44.1%
English → Bambara	39	15	38.5%
English → Portuguese	51	13	25.5%

What the outliers look like

The high-rate outliers are mid-volume, lower-supply languages where a single missing interpreter on shift creates cliff-edge cancellation rates. These are also strong candidates for cross-language back-up arrangements (e.g. Sylheti pooled with Bengali; Cantonese with Mandarin).

3.2 Demand vs unmet — top 10 pairs

4 Interpreter Sizing

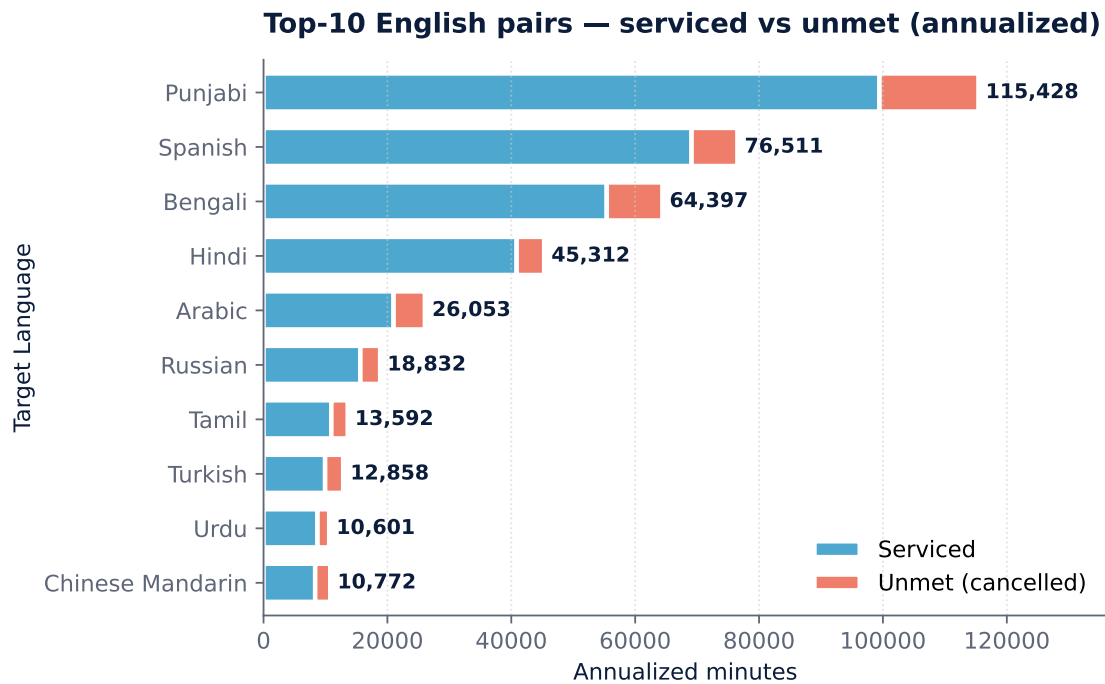


Figure 5: Annualized demand for the top-10 English pairs. Punjabi shows the largest absolute unmet volume but the rate (13.8%) is below average; the unmet bar simply scales with volume.

4.1 Top-10 dedicated head-count requirement

Pair	Min/mo (true)	Cons. (4.8k)	Std (6.4k)	Aggr. (8.0k)
English → Punjabi	9,619	3	2	2
English → Spanish	6,372	2	1	1
English → Bengali	5,385	2	1	1
English → Hindi	3,776	1	1	1
English → Arabic	1,980	1	1	1
English → Russian	1,571	1	1	1
English → Tamil	1,102	1	1	1
English → Turkish	1,120	1	1	1
English → Urdu	876	1	1	1
English → Chinese Mandarin	866	1	1	1
Top-10 subtotal	32,667	14	11	10
Long-tail (66 other languages)	8,401	22–40	≈ 22	≈ 18
Total (per-pair)	41,068		77	70

4.2 Shared-pool reading

If interpreters are treated as fungible across languages they cover (realistic for a centralised on-demand model), the same total demand collapses to:

- **Standard capacity (6,400 min/mo): 7 interpreters**
- **Conservative capacity (4,800 min/mo): 9 interpreters**

- Aggressive capacity (8,000 min/mo): 6 interpreters

4.3 Peak concurrency

The busiest single hour in the four-month window saw **12 simultaneous English-source calls in progress**; the 95th-percentile busy hour was **7**. Peak-hour staffing of **8–10 interpreters available 9:00–14:00 EST** would cover concurrency even on outlier days.

The headcount question, summarised

At a steady-state, weighted-average load this client requires **7–11 interpreters** of English-capable supply. Spread across shifts and accounting for peak concurrency, a working **12–15-interpreter roster** with depth in the top-10 languages delivers a 95%-fill experience at this client's observed volume.

5 Comparison with French-Source Demand

Metric	English-source	French-source
Calls (4 mo)	7,288	337
Cancellation rate	15.9%	75.7%
Distinct target languages	89	4
Annualized true demand (min)	492,814	41,388
Interpreters needed (shared pool)	7	1
Peak concurrency	12	2

The two reports are complementary. The English picture is one of *healthy operations* with a few language-specific gaps and a notable video-modality reliability problem. The French picture is one of *systemic supply failure*, almost entirely concentrated in French → Spanish.

6 Recommended Next Actions

1. **Diagnose the video cancellation gap.** Audio cancels at 4.6%, video at 30.7%. Pull a sample of 50 cancelled video calls and classify root cause (interpreter unavailable, video connect failure, caller hang-up). The fix here is likely operational, not capacity.
2. **Right-size the top-10 pool first.** The standard-capacity sizing gives you a 11-interpreter dedicated team covering 80% of demand. Build that, then layer the long tail on top as a shared on-demand network.
3. **Pair underperforming small-volume languages with their cousins.** Sylheti ↔ Bengali; Cantonese ↔ Mandarin; Moroccan Arabic ↔ Modern Standard Arabic — explicit fall-back routing would likely halve the outlier cancellation rates.
4. **Re-baseline against May–Dec 2025 actuals** when the full year is available; replace the ×3.0 projection.
5. **Add cross-client roll-up** once equivalent Voyce exports for the rest of the SGP network are received.

Appendix — Companion Workbook

The full data and pivots used to generate this report are bundled in `reports/data/Voyce_English_Analysis.xls` with the following tabs:

Tab	Contents
1. Summary	KPIs and headline numbers
2. English Rows (raw)	All 7,288 source-language = English rows
3. Demand by Pair	Serviced minutes / pair across all 76 languages, annualized
4. Cancellation Audit	Cancellation by pair, hour, weekday
5. True Demand	Serviced + imputed-unmet minutes
6. Interpreter Sizing	Per-pair head-count under three capacity scenarios